Monte Carlo Tree Search Exercises

Question 1:

Given a parent node with value 10 and its child nodes with values [12, 7, 15], explain which node should be selected next under a Monte Carlo Search approach.

Answer:
In basic Monte Carlo Search, the selection depends on **simulating random rollouts** from each child and estimating expected rewards. Assuming we consider the current values as estimates of expected reward:

- Child values: 12, 7, 15
- Highest expected reward = 15
  ✅ Therefore, the next action should select the child node with value **15**.

Question 2:

If a parent node has been visited 5 times and a child node has been visited 2 times with a value of 8, how does Monte Carlo Tree Search (MCTS) balance exploration vs. exploitation when choosing the next action?

Answer:

MCTS uses **Upper Confidence Bound for Trees (UCT)**:

$$UCT = \text{Value} + C\sqrt{\frac{\ln(\text{parent visits})}{\text{child visits}}}$$

- Exploitation: favors nodes with high current value (8).
- Exploration: favors nodes visited fewer times.
- Calculation:

$$UCT = 8 + C\sqrt{\frac{\ln 5}{2}}$$

- The child with the highest UCT score will be selected next, balancing **value and uncertainty**.

Question 3:

Given a parent node with multiple child nodes, each with a value and visit count, explain how you would choose the next action in Monte Carlo Search without performing full rollouts.

Answer:

- Use **estimated node values** or **UCT formula** (if using MCTS).
- Compare child nodes based on **current value + exploration bonus**.
- Select the child with **highest combined score** as the next action.
- This allows the algorithm to focus on promising nodes while occasionally exploring less-visited nodes.

Question 4 (Applied Example):

Parent node visits = 10, children nodes:

Child	Value	Visits
A	5	2
B	8	4
C	7	1

Using UCT with $C = 1$, which node should be selected next?

Answer:

UCT formula: $UCT = \text{Value} + C\sqrt{\ln(\text{parent visits})/\text{child visits}}$

- Child A: $5 + \sqrt{\ln 10/2} \approx 5 + 1.07 = 6.07$
- Child B: $8 + \sqrt{\ln 10/4} \approx 8 + 0.76 = 8.76$
- Child C: $7 + \sqrt{\ln 10/1} \approx 7 + 1.52 = 8.52$

✅ **Next action:** select **Child B** (UCT highest).

Here's a **set of additional Monte Carlo Search / MCTS questions** focused on **current node value, parent node info, and next-action selection**, along with answers. These cover both theory and applied examples:

Question 5:

A parent node has been visited 20 times. Two child nodes have the following info:

Child	Value	Visits
X	6	5
Y	9	10

Using the **UCT formula**, which child is more likely to be selected for the next action? Assume (C = 1).

Answer:

UCT = Value + $C\sqrt{\ln(\text{parent visits})/\text{child visits}}$

- Child X: $6 + \sqrt{\ln 20/5} \approx 6 + 0.76 = 6.76$
- Child Y: $9 + \sqrt{\ln 20/10} \approx 9 + 0.54 = 9.54$

✅ **Next action:** Child Y (highest UCT score).

Question 5 (Conceptual):

Explain why Monte Carlo Search might sometimes choose a child node with a **lower current value** as the next action.

Answer:

- To **explore less-visited nodes** and avoid local optima.
- MCTS uses the **exploration term** (e.g., $\sqrt{\ln \text{parent visits}/\text{child visits}}$) which boosts the score of nodes with fewer visits.
- This ensures **balance between exploitation (high value) and exploration (uncertain nodes)**.

Question 6 (Applied Example):

A parent node has **4 children**, with the following values and visits:

Child	Value	Visits
A	10	8
B	7	3
C	9	5
D	5	1

Question: Using UCT ((C=1.5)), which node should be selected next?

Answer:

UCT = Value + $C\sqrt{\ln(\text{parent visits})/\text{child visits}}$

Assume parent visits = 10

- A: $10 + 1.5 * \sqrt{\ln 10/8} \approx 10 + 0.80 = 10.80$
- B: $7 + 1.5 * \sqrt{\ln 10/3} \approx 7 + 1.55 = 8.55$
- C: $9 + 1.5 * \sqrt{\ln 10/5} \approx 9 + 1.06 = 10.06$
- D: $5 + 1.5 * \sqrt{\ln 10/1} \approx 5 + 2.34 = 7.34$

✅ **Next action:** Node A (highest UCT score).

Question 7 (Conceptual):

How does the **number of parent visits** affect the selection of child nodes in Monte Carlo Tree Search?

Answer:

- Higher parent visits increase the **logarithmic term** in UCT.
- This **increases the exploration bonus** for less-visited child nodes.
- Helps the search **shift focus gradually** from exploring to exploiting the best-performing nodes.

Question 8 (Scenario-Based):

A robot using Monte Carlo Search has the following child nodes:

Child	Value	Visits
P	8	2
Q	6	1
R	9	5

If the robot performs **another simulation from child Q**, explain how this may affect the **next action choice** in subsequent iterations.

Answer:

- Additional simulation increases **Q's visit count** and updates its value estimate.
- UCT score may **increase or decrease** depending on the new value.
- This allows Monte Carlo Search to **update beliefs dynamically** and potentially choose Q if it shows high reward after exploration.